

according to such parameters as time and region. Net-metering enables buildings that have solar panels installed to submit excess energy generated to discoms to earn money, or store for later use. These statistics can also be monitored in real time by discoms as well as customers using IoT-enabled solutions.

India's minister of power, R.K. Singh said at a recent event, "It is important to improve India's net-metering facility alongside the adoption of smart meters to propel the renewable energy ecosystem of the country. We have set ourselves a target of converting 100 per cent of energy meters in India into smart meters in the coming three years. This will complement our target of installing 175GW of renewable energy by 2022 and enable the nation to adopt solar energy rapidly."

Artificial intelligence (AI) and data analytics technologies are also playing an interesting role in digitising renewable energy. Once thought to be completely unrelated with renewables, AI-powered software can crunch through historical and real-time energy statistics that are recorded through sensors, and provide predictions for future power generation and consumption trends. It can also notify about power pilferage.

The IoT combined with AI platforms can be used by utilities for infrastructure maintenance, too, to have clear visibility of defects in the grid or

power assets on the ground, and to take immediate action. In short, these modern technologies are kicking in predictive maintenance in utilities.

Opportunities for solution providers

The energy sector has great potential for renewables equipment and solution businesses as well as digital technology solution providers. A recent report by Allied Market Research states that the global solar energy market is set to grow to US\$ 422 billion by 2022, with a CAGR of 24.2 per cent (2016-2022). Given India's ambitious renewable targets of 175GW renewable capacity (including solar, wind and biomass) by 2022 and noticing that barely 74GW has been installed so far, demand for renewable solutions will remain at its peak, especially in the next four years.

The India Brand Equity Foundation (IBEF) reports that India's electricity demand will shoot up to 15,280TWh by 2040. This demand can only be quenched through a strong renewable ecosystem, and this shows us a long-standing business ecosystem. Since India needs to maintain the infrastructure and emphasise on reduction of energy wastage and theft, smart solutions will be in high demand, too.

The smart meter business sector is already quite competitive. Teaming

up with Energy Efficiency Services Ltd (EESL), New Delhi Municipal Council (NDMC) has become the first discom to have converted all meters under their area into smart meters. Connectivity of smart solutions using cellular, Wi-Fi and low-power WAN technologies will also drive the demand for telecom and connectivity businesses, simultaneously.

In short, equipment manufacturers, software providers, system integrators, communication partners, distributors and supply partners in the IoT and renewable ecosystem are looking at strong business opportunities.

The concerns

Both renewable and digital technologies have their own separate set of challenges that need to be addressed. From the renewable side, proper technical installation and maintenance need to be attended to carefully.

Regarding digital technologies, finding the right solutions partner, proper on-field deployment and upgradation of legacy systems, ensuring their compatibility with digital platforms, will not be easy tasks. The initiatives will be capital-intensive and, hence, will require steady support from the government.

Deeper analyses will reveal granular challenges. However, longstanding benefits will be much more emphatic than immediate hardships.

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SMART PARKING:

"It SHOULD BE Like The TELECOM LANDSCAPE, Where There Is INTEROPERABILITY"

- Smart parking solutions can help parking lot owners increase business revenues in many ways. Commuters can also avoid the tiring tales of finding a parking spot. Paromik Chakraborty of *Electronics For You* speaks with Udaya Bhaskar Rao Abburu, chief executive officer and managing director, iRAM Technologies, to understand the benefits and limitations of the smart parking technology

Q. How can smart parking solutions benefit users and parking lot owners, simultaneously?

A. Main constituents of a smart parking solutions are wireless sensors,

gateways, repeaters and the cloud. Sensors are installed on the road or parking area. These communicate with gateways to transmit data, which gets stored in the cloud.

Software and mobile apps are provided to parking owners as well as commuters, who can go online to check the availability of parking spaces beforehand, pre-book parking